

Project Proposal

Group 30

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1 Overview

This project focuses on segmenting adult gliomas from MRI scans using a 2D U-Net convolutional neural network. The BraTS 2021 dataset provides four co-registered MRI sequences for each subject: T1, T1CE, T2, and T2-FLAIR, together with an expert-annotated 3D tumor segmentation mask. Our model will operate on 2D axial slices extracted from these volumes. For each slice, the corresponding images from all four MRI sequences are stacked as input channels, and the model predicts which image regions correspond to background or to one of the tumor sub-regions defined by BraTS.

Manual tumor segmentation requires trained raters to delineate tumor regions throughout a 3D MRI volume and is therefore time-consuming. An automated model can generate an initial segmentation that can be reviewed and corrected by an expert, reducing the amount of manual delineation required. Automated segmentation is also useful for quantitative research because a fixed model applies the same learned segmentation procedure to every scan, providing more repeatable measurements across large datasets than repeated manual segmentation. This can make it easier to compare tumor regions across subjects or across scans collected at different time points.

Brain tumor segmentation is challenging because gliomas vary substantially between subjects in size, shape, location, and appearance. Tumor tissue can also resemble surrounding healthy tissue, and tumor regions may occupy only a small fraction of an MRI slice, creating a strong imbalance between tumor and background. In addition, different tumor sub-regions are distinguished by patterns across T1, T1CE, T2, and T2-FLAIR rather than by a single MRI sequence alone. The model must therefore combine information from the four aligned inputs while identifying irregular tumor boundaries and separating tissue classes with similar appearances.

2 Task Definition

2.1 Machine Learning Task

2.2 Model Input

2.3 Model Output and Segmentation Classes

2.4 Technical Challenges

3 Solution

U-Net is a convolutional neural network (CNN) architecture designed for image segmentation tasks. This architecture consists of an encoder and decoder. The encoder repeatedly applies convolutions and downsamples the image. It learns increasingly abstract features. The decoder upsamples these features back up to the original resolution of the input. Its job is to figure out the exact location of features. In the context of our project, encoded early layers will detect edges and features, while deeper layers can learn patterns associated with tumors. [1]

Each MRI scan consists of four images: T1, contrast-enhanced T1 (T1CE), T2, and T2-FLAIR. These images will be stacked as separate input channels, allowing the model to use information from each image concurrently in its prediction.

The target labels will consist of the already labeled segmentation masks provided with the dataset. Therefore, as the labels are pixel level, the model will also perform pixel-wise classification, assigning each pixel in the U-Net slice to either the background or one of the annotated tumor regions.

The model will be built based off a U-Net architecture, which is a tried and proven workflow designed specifically for biomedical image classification tasks. PyTorch will be used for implementation and training/validation.

References

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. *CoRR*, abs/1505.04597, 2015.